

I. Axioms of Probability

Sample Space and Events

Random Experiment:

A random experiment is a procedure whose outcome is uncertain (or unpredictable).

Examples: Tossing a die or coin

- Drawing a card from a deck
- Tomorrow's temperature/weather

Sample Space and Events:

• The set of all possible outcomes of a random experiment is called the sample space and is usually denoted by S .

The outcomes are sometimes called sample points (or points) in S .

- Subsets of sample space S are called events and are usually denoted by A, B, C, D , etc. E.g., $A \subset S$.

Examples:

(1) Flipping a coin: $S = \{H, T\}$

(2) Rolling a die: $S = \{1, 2, 3, 4, 5, 6\}$

Event A : "outcome is even", then $A = \{2, 4, 6\}$

outcome
is Heads
↓
outcome
is Tails
↓

(3) Flip a coin repeatedly until obtaining a total of two Heads or a total of two Tails:

Here sample space: $S = \{HH, TT, THT, THH, HTH, HTT\}$

↑
outcomes are variable-length sequences

The event A "two flips are needed to stop" is

$$A = \{HH, TT\}$$

The event B "three flips are needed to stop" is

$$B = A^c = \{THT, THH, HTH, HTT\}$$

(4) A coin is flipped 3 times and the sequence of outcomes is recorded. So sample space is

$$S_1 = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

But now suppose instead that the number of Heads is recorded. Here, sample space is

$$S_2 = \{0, 1, 2, 3\}$$

Note: Given a random experiment, a sample space depends on what outcomes are being observed and recorded (so more than one sample space can be formed from a single experiment)

(5) A patient arrives to an 9:00 AM dentist appointment no later than 9:30 AM. The amount of time (in minutes) the patient is late is observed and recorded. Here sample space is

$$S = \{t \in \mathbb{R} : 0 \leq t \leq 30\} = [0, 30].$$

The event A "the patient arrives more than 10 minutes late" is $A = \{t \in S : 10 < t \leq 30\} = (10, 30]$.

$$A \subseteq S \leftarrow$$

(6) Observe and measure the lifetime of a bulb in hours. Here

$$S = \{t \in \mathbb{R} : t \geq 0\} = [0, \infty) = \mathbb{R}^+$$

• The subset

$$B = [1000, \infty)$$

is the event that "the bulb lasts at least 1000 hrs"

• Also the subset

$$C = \{25.7\}$$

is the event that "the bulb lives exactly 25.7 hrs."

Note:

A sample space need not be finite in general.

Nomenclature of Probability Theory

In probability theory, a slightly different terminology is used than in set theory:

Set theory

Universal set S
(universe)
subset E

Probab. theory

$\longleftrightarrow$ Sample space S
 $\longleftrightarrow$ event E

• The sample space S is called the "certain event" since its occurrence is inevitable (as it contains all the outcomes)

• The empty set $\emptyset = S^c$ is called the "impossible or null event"

- Given $E \subset S, F \subset S$, if the outcome of the random experiment belongs to E , we say that "event E occurs"

Thus:

$$E^c \longleftrightarrow \text{"E does not occur"}$$

$$E \cup F \longleftrightarrow \text{"at least one of E or F occurs"}$$

$$E \cap F \rightarrow EF \longleftrightarrow \text{"both E and F occur simultaneously"}$$

$$E - F \longleftrightarrow \text{"E occurs but F does not occur"}$$

- If $E \subset F$, then F occurs whenever E occurs: We say "E implies F"
 $\uparrow$
 $\text{or } \subseteq$

• Given events $E_1, E_2, \dots, E_n$, then

$$\bigcup_{i=1}^n E_i = \{x \in S : x \in E_i \text{ for some } i=1, \dots, n\}$$

is the event that "at least one of E_i 's occurs"

and

$$\bigcap_{i=1}^n E_i = \{x \in S : x \in E_i \text{ for all } i=1, \dots, n\}$$

is the event that "all E_i 's occur simultaneously".

E.g., If events $E_1, E_2, \dots, E_n$ are mutually exclusive, i.e.,

$$E_i \cap E_j \rightarrow E_i E_j = \emptyset \quad \forall i \neq j,$$

then

$$\bigcap_{i=1}^n E_i = \emptyset \quad (\text{impossible event})$$

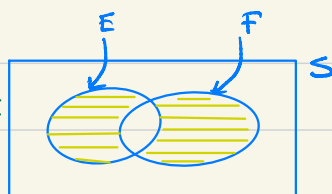
Example:

(a) Given events E and F in sample space S , the event "exactly one of E or F occurs" is given by

$$(E - F) \cup (F - E).$$

Let us show via set operation properties that

$$(E - F) \cup (F - E) = (E \cup F) - EF$$



Indeed, have

$$\begin{aligned} (E - F) \cup (F - E) &= (EF^c) \cup (FE^c) && \text{by def of set difference} \\ &= (E \cap F^c \cup F \cap E^c) && \text{by distributivity of } \cup \text{ over } \cap \\ &= (E \cup F) (\underbrace{F^c \cup F}_= S) (\underbrace{E^c \cap E}_= S) && \text{"} \\ &= (E \cup F) S S (F^c \cap E^c) \\ &= (E \cup F) (F^c \cap E^c) && \text{de Morgan's law} \\ &= (E \cup F) ((EF)^c) \\ &= (E \cup F) - EF \quad \square \end{aligned}$$

(b) Given events E, F, G in sample space S , the event "at least one of E, F or G occurs" is given by

$$E \cup F \cup G$$

and

the event "at most one of E, F, G occurs" is given by

$$\underbrace{(E \cup F \cup G)^c}_{\text{none of them occurs}} \cup \underbrace{[(E \cup F \cup G) - ((E \cap F) \cup (E \cap G) \cup (F \cap G))]}_{\text{exactly one of them occurs}} \quad (b, c)$$

$$= (E \cup F \cup G)^c \cup \overline{[(E \cap F) \cup (E \cap G) \cup (F \cap G)]}$$

Example: A die is repeatedly rolled until a 6 occurs.

(a) Let us describe the sample space S of the experiment.

S consists of all finite-length sequences of integers between 1 and 6 such that 6 appears only in the last position plus all infinite-length sequences in which 6 does not appear:

$$S = \left(\bigcup_{n=1}^{\infty} A_n \right) \cup B$$

where

Event: " n rolls are needed to complete experiment"

$$A_n = \left\{ \underbrace{a_1, a_2, \dots, a_{n-1}}_{\text{length-}n \text{ sequence with 6 only appearing in last position}}, 6 : 1 \leq a_i \leq 5, i=1, \dots, n-1 \right\}$$

and

$$B = \left\{ \underbrace{a_1, a_2, \dots}_{\text{infinite-length sequence where 6 never appears}} : 1 \leq a_j \leq 5, j=1, 2, \dots \right\}$$

(b) What is the event $\left[\bigcup_{n=1}^{\infty} A_n\right]^c$?

This is indeed the event "6 never appears":

$$\begin{aligned}\left[\bigcup_{n=1}^{\infty} A_n\right]^c &= \left\{ a_1 a_2 a_3 \dots : 1 \leq a_j \leq 5, j=1, 2, \dots \right\} \\ &= B.\end{aligned}$$
